

CSC360 - Computer Graphics and Digital Image Processing

25/8/26 Reflection

Topic: Data Structures, XML, Threads and JavaFX

Today, in class, I have studied various topics of Java Programming and Graphical Applications and also Project Structures. We have also been made to understand that tree data structures represent relations between different nodes, and these concepts can also be studied from the perspective of parents and children.

Points to Remember

Tree Data Structure:

A tree has parent nodes and child nodes with a hierarchical connection between them.

- Different types of trees may include both vertical and horizontal ones.
- Parents and children show connections between different elements.
- This structure will be helpful while working with files, folders, and XML.

Project and XML Files

We have learned some common files used in a Java project like pom.xml, .gitignore, target, int, out, and .idea.

- pom.xml → handles configuration and dependencies in a Maven project.
- .gitignore → instructs Git to ignore specific files/folders.
- target / out → stores generated/compiled files of a project.
- .idea → contains project settings specific to an IDE.

Processes and Threads

In this lecture, we learned about the distinction between process and threads. Process refers to the currently executing program whereas threads are smaller sub-units that run inside a process.

Even when a program does not perform multitasking, there can still be situations like in a GUI application, where it is necessary for threads to be used.

Thread Safety

The need for thread safety when several threads access the same data was discussed in this lecture. Thread safe functions or data structures require locks, mutexes or synchronization, all of which tend to make programs more complex and slower.

Thus, if it is not necessary to have concurrent access, it is preferred to use a non-thread safe method since it is faster and simpler.

JavaFX

Finally, we discussed how to run a JavaFX application, including the steps on how to include JavaFX dependencies and launch the main JavaFX class file from the configuration of our project.

Key Learning

In today's lecture, I found out how data structures, project files, XML, processes, threads, and JavaFX are related to one another. They are not just concepts but tools that help us manage our projects and data and even understand how Java code is executed.